

Part 6: The Classical String Model and the Lund Model

How a Meson Is a Yo-Yo, and How Strings Break into Hadrons

PHYS 7502 – Final Reading Assignment

1 Where We Are

In Part 5 we built the physical picture: the color field between a q and $\bar{q}$ is confined to a flux tube with linear energy, $U(L) = \kappa L$. When the tube stretches far enough, it breaks by Schwinger-type pair production, releasing mesons. This is the basic mechanism of hadronization.

Now we're going to turn that picture into an actual *mechanical model* – one you can write down equations for, solve, and use for predictions. That's the **classical string model**, and its specific implementation that underlies PYTHIA is the **Lund model**.

2 Historical Route: The Dual Resonance Model

Before we dive into the mechanics, a short historical note. The string model of hadrons *predates* QCD – it grew out of the **dual resonance model** of the late 1960s.

2.1 The two ways to compute a scattering amplitude

In studying meson-meson scattering at high energies, people noticed something peculiar. You could compute the amplitude two ways:

- **s -channel**: via formation of intermediate resonances that decay into the final state.
- **t -channel**: via exchange of particles carrying the momentum transfer.

At low energies, these two pictures give visibly different amplitudes. But at high energies, the s -channel amplitude (summed over all intermediate resonances) turned out to *equal* the t -channel amplitude (summed over all exchanges).

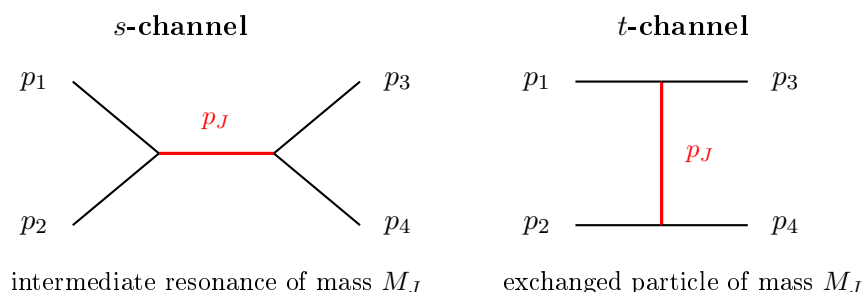


Figure 1: The two diagrams for a meson-meson elastic scattering amplitude. *Duality* says that the two summed over all intermediate/exchanged particles give the same answer. The interpretation that makes duality natural: the mesons are actually **strings**.

Duality

The equality of the s - and t -channel amplitudes at high energy is called **duality**. The simplest way to understand it geometrically: if mesons are strings rather than point particles, the two diagrams have the same topology (just rotated 90 degrees). So of course they give the same amplitude! This is the origin of the name *dual resonance model*, which was the precursor to modern string theory.

2.2 Regge trajectories

A remarkable empirical observation fell out of this work. When you plot meson (or baryon) states with *same internal quantum numbers* but different spins J on a graph of J versus M^2 , you get **straight lines**:

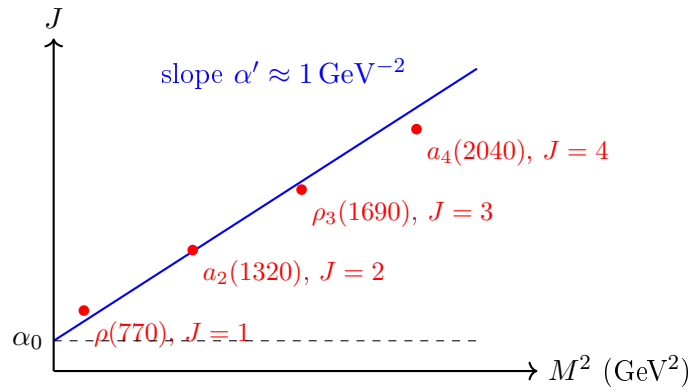


Figure 2: A typical Regge trajectory: plotting hadron spin J vs. mass-squared M^2 yields a straight line. The intercept is α_0 , the slope is α' . For light mesons (like the ρ family shown), $\alpha' \approx 1 \text{ GeV}^{-2}$.

Regge Trajectory

States on a Regge trajectory satisfy:

$$J(M) = \alpha_0 + \alpha' M^2 \quad (1)$$

where α_0 is the **Regge intercept** and α' the **Regge slope**. Empirically, for mesons and baryons,

$$\alpha' \approx 1 \text{ GeV}^{-2} \quad (2)$$

independent of the quantum numbers of the trajectory.

Why should J be linearly related to M^2 ? It's not obvious from any point-particle picture. But...

3 The String Derivation of the Regge Trajectory

Here's where the string picture pays off. Consider a meson as a relativistic string of length $2L$, rotating as a rigid body about its center. The endpoints (the q and $\bar{q}$) travel at the speed of light.

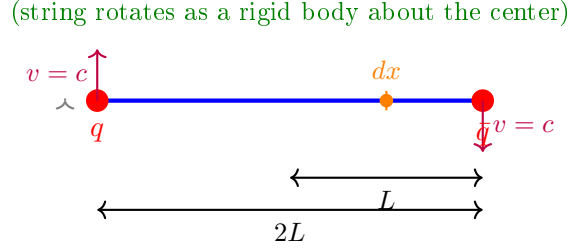


Figure 3: The rotating relativistic string model of a meson. Quarks at the endpoints travel at the speed of light, in opposite directions (one up, one down, as the string rotates). A string element (orange) at distance x from the center moves at velocity $v(x) = (x/L)c$, growing linearly from zero at the center to c at the endpoints.

3.1 Computing M and J

Each element dx of the string has energy density κdx (at rest). In the rotating frame, an element at distance x from the center moves at $v(x) = (x/L)c$, so its gamma factor is $\gamma(x) = 1/\sqrt{1 - (x/L)^2}$.

Total mass. Integrating the boosted energy density over the string:

$$\begin{aligned} M &= 2 \int_0^L \gamma \kappa dx = 2\kappa \int_0^L \frac{dx}{\sqrt{1 - (x/L)^2}} \\ &= 2\kappa \cdot L \cdot \frac{\pi}{2} = \pi\kappa L \end{aligned} \quad (3)$$

Total angular momentum. Each element contributes $\gamma\kappa v x dx$ (angular momentum = energy $\times$ velocity $\times$ distance, in natural units):

$$\begin{aligned} J &= 2 \int_0^L \gamma \kappa \cdot (x/L) \cdot x dx = \frac{2\kappa}{L} \int_0^L \frac{x^2 dx}{\sqrt{1 - (x/L)^2}} \\ &= \kappa L^2 \cdot \frac{\pi}{2} \end{aligned} \quad (4)$$

Eliminate L . From (3), $L = M/(\pi\kappa)$. Substitute into (4):

$$J = \kappa \cdot \frac{M^2}{\pi^2 \kappa^2} \cdot \frac{\pi}{2} = \frac{M^2}{2\pi\kappa} \quad (5)$$

The String Prediction for Regge Slope

$$J = \frac{1}{2\pi\kappa} M^2 \quad (6)$$

This has exactly the form of the empirical Regge trajectory, equation (1), with intercept $\alpha_0 = 0$ and slope

$$\alpha' = \frac{1}{2\pi\kappa} \quad (7)$$

Using the measured $\alpha' \approx 1 \text{ GeV}^{-2}$, we infer

$$\kappa = \frac{1}{2\pi\alpha'} \approx \frac{1}{2\pi \cdot 1 \text{ GeV}^{-2}} \approx 0.16 \text{ GeV}^2 \approx 1 \text{ GeV/fm} \quad (8)$$

This is beautiful. A simple classical picture – a rotating relativistic string with tension κ – *quantitatively* predicts the observed Regge slope. This was already suggestive evidence for the string picture in the 1970s, before QCD was fully developed.

String Theory Disclaimer

Although strings describe hadrons' large-distance structure beautifully, *fundamental* string theory has problems as a theory of strong interactions (it predicts a massless spin-2 particle, the “graviton”, which doesn't exist as a meson). That's why string theory was later repurposed as a theory of *quantum gravity*. The classical string model of hadrons is an **effective** description, valid at hadronic length scales but not fundamental.

4 The Yo-Yo Model: A Meson at Rest

Now we'll build up the simplest dynamical string model, the **yo-yo model**. The picture: a meson at rest in its center-of-mass frame is a straight string with a q at one end and a $\bar{q}$ at the other, with the endpoints oscillating back and forth.

4.1 The Hamiltonian

For simplicity, assume the quark and antiquark are massless. The string is a 1D object with tension κ , so the potential energy is $\kappa|x_q - x_{\bar{q}}|$. The kinetic energy is relativistic (massless particles): $|p|$ for each.

$$H(x_i, p_i) = |p_q| + |p_{\bar{q}}| + \kappa|x_q - x_{\bar{q}}| \quad (9)$$

This is the Hamiltonian for the yo-yo system.

4.2 Solving the motion

The equations of motion are Hamilton's equations applied to (9):

$$\dot{x}_i = \frac{\partial H}{\partial p_i} = \text{sgn}(p_i) \quad (10)$$

$$\dot{p}_i = -\frac{\partial H}{\partial x_i} = -\text{sgn}(x_i - x_{i'})\kappa \quad (11)$$

Several things are notable:

- **Velocity is $\pm c$.** Since quarks are massless, they always move at the speed of light. The sign of the momentum determines the direction.
- **Momentum changes at rate $\pm\kappa$.** As the quarks move apart, each feels a pull from the string of magnitude κ , directed toward the other.

4.3 The yo-yo motion

Starting from $x_q = x_{\bar{q}} = 0$ with $p_q(0) = p > 0$ and $p_{\bar{q}}(0) = -p$:

1. The quark moves to the right at speed c with decreasing momentum.
2. At time $T/4 = p/\kappa$, its momentum hits zero. It has traveled a distance p/κ .
3. Now the string pulls it back. Momentum goes negative, so velocity is $-c$: it moves back toward the center.
4. At $T/2$, the quark returns to $x = 0$ with momentum $-p$. The antiquark does the mirror-image motion.
5. The pattern repeats forever – a yo-yo oscillation.

Yo-Yo Period

$$T = \frac{4p_q(0)}{\kappa} \quad (12)$$

For a ρ meson with $M \approx 770$ MeV, total energy $= 2|p_q| \approx 770$ MeV, so $|p_q| \approx 385$ MeV. Thus $T \approx 1.54 \text{ fm}/c \approx 5 \times 10^{-24} \text{ s}$ – a very fast oscillation, characteristic of hadronic scales.

4.4 Space-time trajectories

A beautifully simple picture emerges when you plot x versus t :

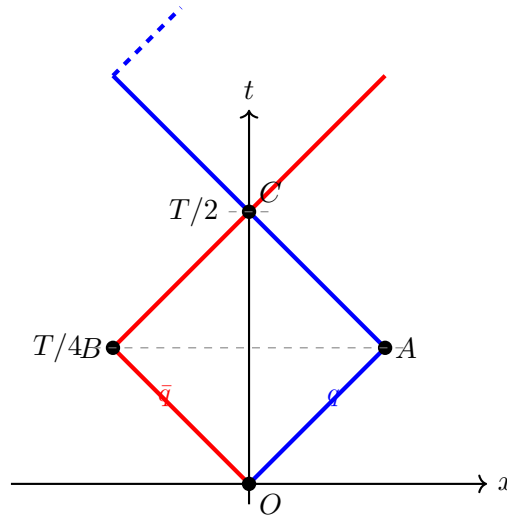


Figure 4: Space-time trajectories of a yo-yo meson. The quark (blue line) moves to the right until time $T/4$ (point A), then reverses and comes back to the origin at $T/2$ (point C). The antiquark (red line) does the mirror-image motion. O , A , B , C are the vertices of a parallelogram; its area turns out to be a Lorentz invariant.

4.5 The key invariant: area equals s/κ^2

Here's the punchline of the yo-yo picture. Introduce **light-cone coordinates**:

$$u = t + x, \quad v = t - x \quad (13)$$

In the (u, v) plane, the four points O, A, B, C of the parallelogram become:

$$O = (0, 0), \quad A = (T/2, 0), \quad B = (0, T/2), \quad C = (T/2, T/2) \quad (14)$$

(a rectangle, now). The area of this rectangle is $(T/2)^2 = T^2/4$. Using $T = 4p_q(0)/\kappa$:

$$\text{area}(OACB) = \frac{4|p_q(0)|^2}{\kappa^2} = \frac{s}{\kappa^2} \quad (15)$$

since the invariant mass squared of the yo-yo is $s = (2p_q(0))^2 = 4p_q^2(0)$ (two massless particles head-on with equal and opposite momenta).

Meson Area Invariant

In the light-cone (u, v) plane, the area enclosed by one period of the yo-yo trajectory is

$$\boxed{\text{Area} = \frac{M^2}{\kappa^2}} \quad (16)$$

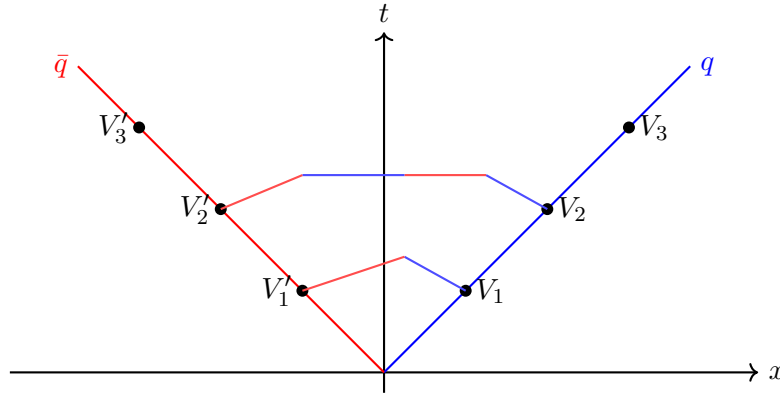
This is a **Lorentz invariant**: the mass M is invariant, and so is the enclosed area. Boost the yo-yo to any speed, and the area of its space-time loop stays the same.

5 The Symmetric Lund Model: How Strings Break

So far we've described a *stable* meson as a yo-yo. Now we handle the case of an excited string that's going to break up into multiple hadrons – the actual hadronization process.

5.1 The fragmenting string

Start with a high-energy $q\bar{q}$ pair produced back-to-back (say, from $e^+e^- \rightarrow q\bar{q}$). As they separate, the string stretches between them. At various space-time points V_i , new $q_i\bar{q}_i$ pairs are produced by the Schwinger mechanism (Part 5). Between two consecutive vertices, the quark from one pair and antiquark from the next combine to form a yo-yo – a hadron.



new $q\bar{q}$ pair creation vertices V_i
each pair of adjacent vertices defines a hadron

Figure 5: String fragmentation in space-time. The original q and $\bar{q}$ move apart at the speed of light along the outermost trajectories. Between them, the string breaks at various vertices V_i , producing new $q\bar{q}$ pairs. Each pair of consecutive break vertices (via a little yo-yo piece) defines a hadron.

5.2 The key assumption of the Lund model

The Lund model makes one central assumption: the break vertices V_i lie approximately on a curve of **constant proper time** $\tau_0 = \sqrt{t^2 - x^2}$:

$$t^2 - x^2 = \tau_0^2 \quad (17)$$

Why Constant Proper Time?

Think about what determines when the string breaks. Locally, the string has a roughly uniform color electric field $\mathcal{E}$, and Schwinger tunneling happens at some rate per unit 4-volume. This rate is *Lorentz invariant*, so the distribution of break events in space-time should be Lorentz invariant too. A curve of constant proper time from the origin is the most natural Lorentz-invariant choice.

5.3 Consequences: the flat rapidity distribution

If break vertices lie on $\tau = \tau_0$, several beautiful things follow. The most important one is that the **rapidity distribution** of produced hadrons is *flat*:

$$\frac{dN}{dy} = \text{const} \quad (18)$$

Why Flat in Rapidity?

Think about two break vertices at proper times τ_0 and slightly later. In the rest frame of a hadron produced between them, the time interval between breaks is some $\Delta\tau$. But rapidity differences are Lorentz invariants, and the Lund setup is Lorentz-invariant by construction. Therefore the *rate* of hadron production per unit rapidity is constant. This is exactly what you see in data: a “rapidity plateau” in the central region of jets, with the plateau height

determined by κ .

5.4 The Lund splitting function

Combining constant proper time with the additional assumption of **left-right symmetry** (the probability distribution doesn't depend on whether you build the hadron chain starting from the left or the right end), Andersson, Gustafson, and collaborators derived the explicit functional form for the string breaking probability.

Let z_+ be the fraction of the *forward light-cone momentum* $p_+ = E + p_z$ that a particular hadron (of mass squared m^2) takes away. The probability that the hadron is produced with that z_+ is:

Lund Symmetric Splitting Function

$$f(z_+, m^2) = N z_+^{-1} (1 - z_+)^a \exp\left(-\frac{bm^2}{z_+}\right) \quad (19)$$

where:

- N is a normalization constant.
- a controls the small- z behavior.
- b controls the mass dependence; it's connected to the Schwinger exponential e^{-bm^2/z_+} .

Reading the Lund Function

This function is implemented inside PYTHIA. The parameters a, b are what experimental collaborations **tune** (remember the “tuning” discussion in Part 3). Typical values from recent tunes are $a \approx 0.3$, $b \approx 0.8 \text{ GeV}^{-2}$.

- The z_+^{-1} factor favors *soft* particles (low z_+).
- The $(1 - z_+)^a$ factor cuts off *hard* particles (near $z_+ = 1$).
- The e^{-bm^2/z_+} factor is the heavy-quark-suppression inheritance from Schwinger – heavy hadrons are exponentially suppressed.

Together these produce a broad peak at moderate z_+ , which is what you see in data.

5.5 How PYTHIA uses this

In practice, PYTHIA implements this through an **iterative** scheme:

1. Start at one end of the string (say, the quark end).
2. Use $f(z_+, m^2)$ to pick the z_+ of the first hadron (sampled randomly).
3. Subtract that hadron's momentum, leaving a shortened string.
4. Repeat for the next hadron, and so on.
5. Stop when the remaining string is too short to produce another hadron.

Independent Fragmentation

This iterative procedure assumes that each hadron is produced *independently* – no correlation between adjacent hadrons beyond what the splitting function enforces. This approximation is what makes PYTHIA fast, but it misses some subtle correlations that more refined treatments try to put back in.

6 Gluons on Strings: Kinks

One more feature of the Lund model that’s worth knowing: when a gluon is radiated from the $q\bar{q}$ pair, the string doesn’t just attach to the quark – it bends *through* the gluon.

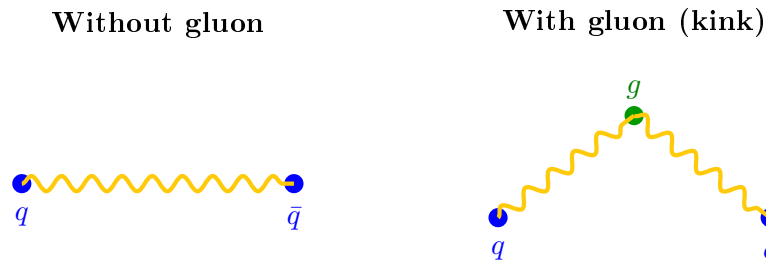
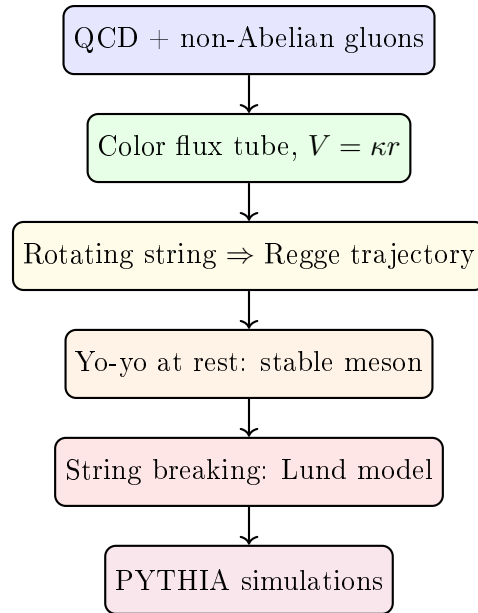


Figure 6: Gluons on strings. A gluon emitted during the shower becomes a “kink” in the string – the string passes through the gluon rather than bypassing it. This is a consequence of the gluon being in the adjoint representation of $SU(3)_{\text{color}}$: it carries both a color and an anti-color, like a string endpoint does.

This prediction, known as the **string effect**, was one of the early successes of the Lund model: it explained asymmetries in 3-jet events at PETRA and LEP that couldn’t be understood with simpler “independent fragmentation” pictures.

7 Tying It All Back

Here’s the compact summary of the logical chain you’ve just been through:



Each step in this chain builds on the one above. QCD plus non-Abelian gauge structure gives flux tubes. Flux tubes give a linear potential. A rotating linear string gives the Regge trajectory. A stable string oscillates as a yo-yo. An energetic string breaks according to the Lund splitting function. PYTHIA packages this all up into concrete simulated events.

8 Summary of Part 6

Main Takeaways

1. The classical string model treats a meson as a relativistic string of tension $\kappa \approx 1 \text{ GeV/fm}$ with quarks at the ends.
2. A rotating relativistic string *naturally predicts* the observed Regge trajectory $J = \alpha_0 + \alpha' M^2$ with $\alpha' = 1/(2\pi\kappa) \approx 1 \text{ GeV}^{-2}$. This was one of the first quantitative successes of the string picture.
3. A meson at rest is modeled as a **yo-yo**: quarks moving at $\pm c$, momentum evolving linearly under the string tension. Period $T = 4p_q(0)/\kappa$.
4. In the (u, v) light-cone plane, the yo-yo's space-time loop has area M^2/κ^2 – a Lorentz invariant.
5. The **Lund model** handles hadronization by assuming string-break vertices lie on a curve of constant proper time. This gives a flat rapidity distribution (observed!) and a specific splitting function

$$f(z_+, m^2) = N z_+^{-1} (1 - z_+)^a e^{-bm^2/z_+}.$$

6. Gluons appear as **kinks** in the string – a prediction confirmed by the string effect in 3-jet events.

7. This entire framework is implemented inside **PYTHIA** and is used daily for LHC physics.